



**Department of Electrical and Computer Engineering
North South University**

Senior Design Project

Optimizing TinyML with Quantization and Distillation for Indoor Localisation in Resource-Constrained Environments

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Fall, 2025

LETTER OF TRANSMITTAL

December, 2025

To

Dr. Mohammad Abdul Matin
Chairman,
Department of Electrical and Computer Engineering
North South University, Dhaka

Subject: Submission of Capstone Project Report on “Optimizing TinyML with quantization and distillation for Indoor Localisation in Resource-Constrained Environments.”

Dear Sir,

With due respect, we would like to submit our **Capstone Project Report** on “**Optimizing TinyML with Quantization and Distillation for Indoor Localisation in Resource-Constrained Environments**” as a part of our BSc program. The report deals with Multi Level Based Home Localization System. This project was very much valuable to us as it helped us gain experience from practical field and apply in real life. We tried to the maximum competence to meet all the dimensions required from this report.

We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative to have an apparent perspective on the issue.

Sincerely Yours,

.....
Mueid Islam Arian
ECE Department
North South University, Bangladesh

.....
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APPROVAL

Mueid Islam Arian (ID # 2121947642), Md. Mahamudul Hasan (ID # 2121379642) and Student Md. Nafiz Un Nabi Ishan (ID # 2121346042) from Electrical and Computer Engineering Department of North South University, have worked on the Senior Design Project titled “Optimizing TinyML with quantization and distillation for Indoor Localisation in Resource-Constrained Environments” under the supervision of Dr. Mohammad Shifat-E-Rabbi partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

Supervisor’s Signature

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Dr. Mohammad Abdul Matin

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' names & Signatures

1. Mueid Islam Arian

2. Md. Mahamudul Hasan

3. Md. Nafiz Un Nabi Ishan

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Furthermore, the authors would like to thank the Department of Electrical and Computer Engineering, North South University, Bangladesh for facilitating the research. The authors would also like to thank their loved ones for their countless sacrifices and continual support.

ABSTRACT

Optimizing TinyML with Quantization and Distillation for Indoor Localisation in Resource-Constrained Environments

Indoor localization has become a key enabler for applications in smart buildings, asset tracking, and context-aware systems. One of the problems with modern state of the art models is how resource heavy and power consuming they are, making them quite difficult to use in resource constrained environments. Deploying deep learning models for accurate localization on resource limited edge devices to this day remains a challenge. In this project, we propose an optimized TinyML framework for indoor localization on edge devices by using model quantization and knowledge distillation to lightweight the architecture for on device indoor localization using the UJIIndoorLoc-Mag dataset, which contains magnetic field measurements for location identification. We developed several lightweight Mamba and Transformer-inspired neural network architecture using LSTM layers as state-space model approximations, optimized specifically for ESP32 microcontroller deployment. Our best models achieve a test F1 score of approximately 75-92% with a quantized model size of only 12.18 KB-27 KB, making it highly suitable for resource-constrained edge devices. This demonstrates the feasibility of deploying sophisticated deep learning models on microcontrollers with limited memory (520KB SRAM) and computational resources. We try to minimize the model size, power consumption and latency while keeping the accuracy as close as to the original models, particularly for healthcare applications like patient monitoring.

Index Terms—TinyML, Indoor Localization, Edge Devices, Quantization, Distillation

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Chapter 1 Introduction

Indoor localization has become an essential component of modern smart environments, enabling applications ranging from assisted living and occupancy monitoring to security systems and energy-efficient automation. While GPS has revolutionized outdoor navigation, its signals degrade significantly indoors, making alternative localization techniques necessary. Among various approaches, Wi-Fi and Bluetooth-based signal-strength analysis have gained attention due to their low cost, ease of deployment, and compatibility with existing infrastructure. However, these systems rely heavily on machine learning and deep learning models that often require substantial computational resources, limiting their feasibility in small, low-power embedded devices.

Recent advancements in Tiny Machine Learning (TinyML) have made it possible to deploy intelligent models on microcontrollers with extremely limited memory and processing capabilities. At the same time, emerging architectures like Transformers and Mamba (Selective State Space Models) have demonstrated strong performance in modeling sequential data. When combined with TinyML techniques such as model optimization, quantization, and knowledge distillation, these architectures can enable accurate indoor localization on inexpensive edge hardware such as the ESP32. This project explores these ideas by building a lightweight indoor localization system using RSSI time-series data and modern deep learning architectures optimized for TinyML deployment.

1.1 Background and Motivation

Indoor tracking and localization systems are widely used in smart homes, healthcare monitoring, industrial automation, and human-activity analysis. Traditional solutions, such as infrared sensors, RFID, or camera-based systems, often suffer from limitations related to cost, privacy, scalability, and environmental dependency. Bluetooth Low Energy (BLE) Received Signal Strength Indicator (RSSI) values provide a non-intrusive and cost-effective way to identify room-level locations based on wireless signal patterns.

However, RSSI data is highly unstable and noisy due to multipath interference, wall thickness, human movement, and device orientation. Classical machine learning models often struggle to capture such temporal variations without extensive feature engineering. Deep learning models like LSTM and Transformers can learn sequential patterns directly from data, but their computational footprint is typically too large for deployment on embedded microcontrollers.

The growing interest in TinyML running neural networks directly on small devices offers a promising opportunity to bring indoor localization systems into practical real-world use. This project is motivated by the need for a low-cost, energy-efficient, privacy-preserving indoor localization system capable of running entirely on a microcontroller without relying on cloud computing. The introduction of efficient architectures like Mamba and optimized Transformer variants, combined with knowledge distillation and quantization, provides a strong foundation for achieving accurate predictions within severe memory limits.

1.2 Purpose and Goal of the Project

The purpose of this project is to design, develop, and evaluate a TinyML-compatible indoor localization model that can operate efficiently on low-power, resource-constrained microcontroller devices. The project aims to construct an indoor localization system using BLE RSSI time-series data gathered from multiple rooms within a residential environment, allowing the model to learn distinctive signal patterns associated with different indoor spaces. In pursuit of improved performance, the system explores and compares several modern sequential learning architectures, including Tiny Transformers, Tiny Mamba (Selective State Space Models), LSTM networks, and lightweight MLP baselines, each selected for their varying capabilities in capturing temporal dynamics and representing complex signal fluctuations. To further enhance efficiency and enable real-world deployment on embedded hardware, the project incorporates key optimization techniques such as knowledge distillation—where a larger Transformer model transfers its learned representations to a smaller Mamba model—and dynamic INT8 quantization to significantly reduce model size and memory requirements without severely compromising accuracy. The final optimized model is then prepared for deployment on TinyML-capable

microcontrollers such as the ESP32, ensuring fully offline operation without reliance on cloud computation. The evaluation of the system involves measuring classification accuracy, F1-score, confusion matrices, and overall model footprint to determine its suitability for continuous indoor tracking applications. At a broader level, this project contributes to the development of affordable smart-home localization solutions that maintain user privacy by avoiding cloud dependency, while also reducing energy consumption, operational cost, and maintenance burdens typically associated with conventional indoor tracking systems. Additionally, it provides a robust foundation for future advancements in smart-home automation, real-time occupancy detection, personalized assistance technologies, and secure environment monitoring, thereby supporting the long-term vision of fully integrated intelligent living spaces.

1.3 Organization of the Report

This report is organized into five chapters, as outlined below:

Chapter 1: Introduction

Introduces the project by providing an overall understanding of its background, motivation, and relevance in the field of indoor localization and TinyML. It outlines the rationale for using BLE RSSI time-series data and explains why lightweight machine learning models are necessary for deployment on low-power microcontrollers. This chapter clearly states the objectives of the research, defines the scope of the work, and emphasizes the importance of developing an accurate yet computationally efficient indoor localization system.

Chapter 2: Literature Review

Presents a comprehensive literature review that examines existing indoor localization approaches, highlighting the strengths and weaknesses of conventional techniques such as Wi-Fi fingerprinting, RFID-based systems, and classical machine learning models.

Chapter 3: Methodology

Outlines the complete methodology adopted throughout the project. It describes the dataset used in the research, including the structure of the RSSI recordings collected from multiple rooms and the reasoning behind choosing time-windowed sequences. The training procedures, evaluation

strategies, knowledge distillation process, and INT8 quantization techniques are discussed thoroughly to demonstrate how the model is optimized for both accuracy and microcontroller compatibility. Furthermore, the metrics used for validation, such as accuracy, F1-score, and confusion matrices, are justified in the context of classification performance.

Chapter 4: Investigation, Experiment, Results, Analysis, and Discussion

This chapter presents a detailed investigation of the proposed indoor localization system, focusing on the experimentation process, model evaluation, and the analysis of results.

Chapter 5: Impacts of the Project

This project has several significant impacts spanning technological, economic, academic, and societal domains. From a technological perspective, it demonstrates that advanced deep learning architectures such as Transformers and Mamba state-space models can be successfully adapted, optimized, and deployed on extremely resource-constrained microcontrollers through TinyML techniques.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

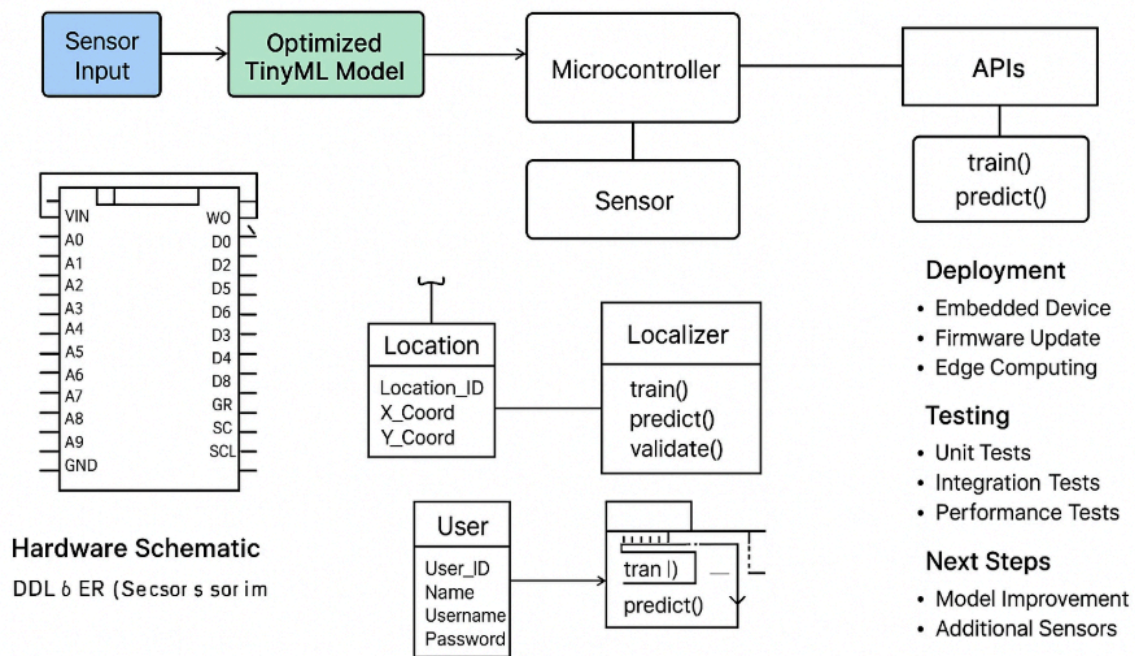
Quantization and knowledge distillation are among the most popular TinyML methods for model compression [4]. Quantization reduces model size and computational cargo by dwindling the perfection of the figures representing the model's parameters, which can incontinently reduce the model's size by half [5]. Its straightforward operation allows for easy implementation, especially when deploying formerly small and effective models on resource-constrained devices. A recent study shows accurate discovery of factory conditions to ensure sustainable agrarian productivity and food security by developing a featherlight, real-time factory complaint discovery model using TinyML, specifically targeting apple and mango leaves [2]. Zeynali, M, addresses the challenge of non-invasive blood glucose position (BGL) monitoring for diabetes operation, aiming to replace traditional cutlet poking styles with a comfortable, nonstop, and accurate volition [3]. Their result, a refined segmentation system, better model delicacy and computational effectiveness. Embedded deployment demonstrated practical connection for nonstop monitoring. A proposal was made for small, practical model families tuned for TinyML benchmark tasks (visual wake words, keyword spotting, anomaly detection) and emphasize reproducible deployment via TFLite Micro [6], allowing reproducible TinyML models that perform well on canonical TinyML tasks using standard quantized/microcontroller toolchains. In knowledge distillation, a compact neural network is optimized to imitate the behaviors of a larger, more complex network. Instead of learning only from ground-truth labels, the student also learns from the soft predictions or intermediate representations produced by the teacher. The goal is to transfer the knowledge from the teacher model to the student model which can run efficiently on microcontrollers. Wang et al. in a recent study showed a fault-diagnosis system for industrial IoT where a large cloud model (teacher) transfers knowledge to a compact edge model (student) [7]. Their goal is to maintain high diagnostic accuracy while enabling real-time inference on low-resource edge devices. One of the key findings of their work is that knowledge distillation (KD) significantly improves the accuracy and robustness of lightweight edge models compared to training them alone.

A lot of these work are done by model contraction ways like quantization, pruning, and knowledge distillation. These are designed to reduce model size; however, each of these come with their own limitations. Moosmann et al. introduced TinyissimoYOLO, an ultra-compact object-detection network optimized through 8-bit quantization for microcontrollers, demonstrating real-time inference and low energy cost, however, its accuracy and class capacity remain limited due to extreme model size reductions [8]. Heydari and Mahmoud conducted a comprehensive TinyML survey, highlighting that while quantization, pruning, and compression enable significant memory and energy reductions, the field still lacks standardized benchmarks across diverse MCU hardware, making cross-study comparisons difficult [9]. Meanwhile, federated learning combined with knowledge distillation has been shown to reduce training and communication overhead on low-resource edge devices, but student models often lack the capacity to fully absorb teacher knowledge and are rarely tested under true TinyML-class memory limits (<100 KB SRAM) [10]. While knowledge distillation (KD) has been proposed as a way to transfer “teacher” knowledge to smaller “student” models suitable for edge, the process faces serious challenges: in constrained settings combining KD with quantization or ultra-small models can amplify error accumulation, and distilled models may still lack the representational capacity to match teacher performance under severe compute/memory constraints. Proposed frameworks for edge-IoT KD often assume relatively capable “edge devices” rather than true microcontrollers, limiting their relevance for tight TinyML targets [11].

Chapter 3 Methodology

3.1 System Design

Block Diagram of the project:



Data Flow & TinyML Pipeline: The system begins with Sensor Input, such as WiFi RSSI, BLE signals, or IMU readings. This raw data is fed into the Optimized TinyML Model, which has been compressed using quantization and distillation to run efficiently on small devices.

The Microcontroller (e.g., ESP32) runs the TinyML inference on-device and processes the sensor data in real time.

Microcontroller & Sensor Operation: The microcontroller interacts with the Sensor module to collect continuous environmental data. The optimized TinyML model runs on this microcontroller to predict indoor location efficiently even in low-resource environments.

APIs (train() and predict()): The system exposes two core API functions.

train() - trains or retrains the localization model

predict() - outputs the predicted indoor location

These APIs may be used in a server environment or testing phase.

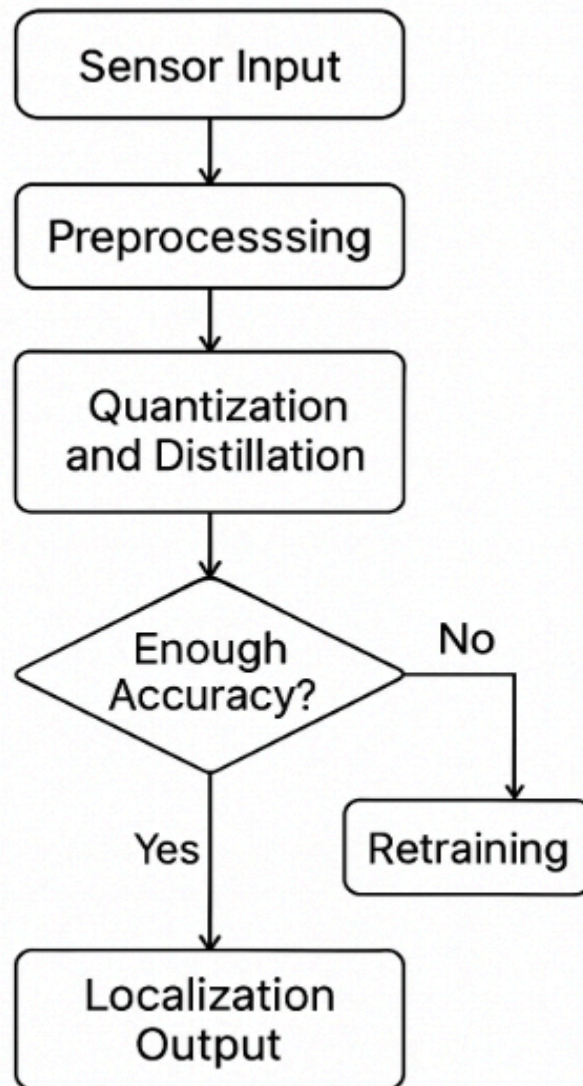
Deployment Section: The diagram outlines where and how the system is deployed:

Embedded Device - the microcontroller running the TinyML model

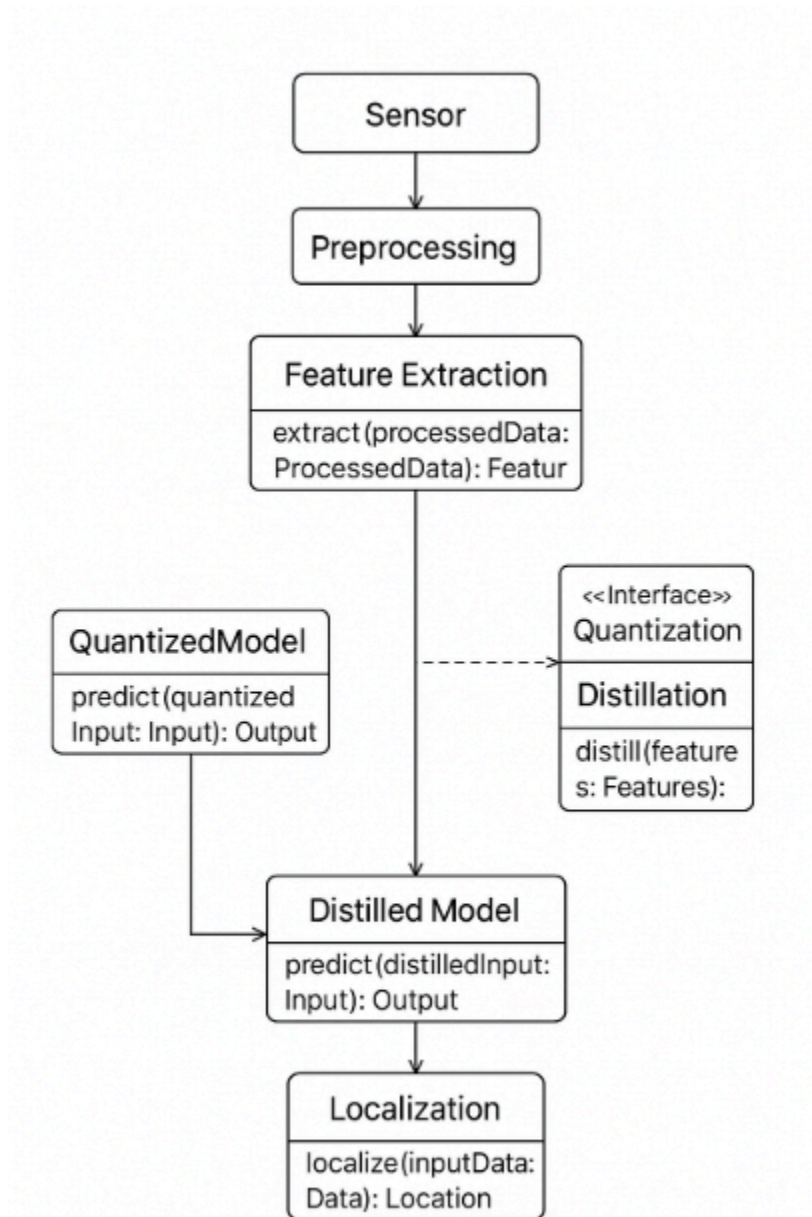
Firmware Update - updates to TinyML model or sensor logic

Edge Computing - all computation occurs locally on the device

Flow chart of the project:



UML diagram of the project:



3.2 Hardware and/or Software Components

Hardware component: The hardware architecture of this project is designed to operate efficiently in resource-constrained environments while supporting reliable indoor localization. The system primarily uses low-power microcontroller such as the ESP32, that serve as the computational platform for running TinyML inference. This device collects and processes data from various sensors including WiFi/BLE RSSI sensors, IMU modules (accelerometer/gyroscope), or optionally UWB modules for higher-precision localization. The hardware also includes a power management module, typically a rechargeable Li-ion battery combined with voltage regulation and sleep-mode optimization to ensure long-term, energy-efficient operation.

Software component: The software system integrates TinyML frameworks, machine learning pipelines, and embedded firmware to create an optimized indoor localization solution. The training environment using Python with TensorFlow, PyTorch, and supporting libraries develops a high-performance teacher model and generates distillation outputs for the lightweight student model. Quantization techniques such as Post-Training Quantization (PTQ) or Quantization-Aware Training (QAT) are applied to produce an ultra-efficient TinyML model deployable on microcontrollers through TensorFlow Lite for Microcontrollers (TFLM). On the device side, embedded C/C++ firmware handles sensor initialization, data preprocessing, feature extraction, and model inference. Additional optional software includes cloud or local database storage, MQTT/Firebase communication, and visualization dashboards for monitoring predicted locations. Together, these software components enable accurate, fast, and memory-efficient localization suitable for low-resource hardware.

3.3 Hardware and/or Software Implementation

Hardware implementation: The hardware implementation focuses on deploying the system on a low-power microcontroller such as an ESP32 or Raspberry Pi Pico, chosen for its affordability and suitability for TinyML applications. Sensors like WiFi/BLE modules or IMU sensors are interfaced with the microcontroller to continuously collect indoor signal data. The device is powered through a compact power management setup typically a Li-ion battery with voltage regulation to ensure stable operation in resource-limited environments. The collected sensor data is processed directly on the microcontroller, enabling real-time indoor localization without depending on external computing resources.

Software implementation: The software implementation combines machine learning optimization techniques with embedded programming to achieve efficient on-device inference. A full-precision teacher model is first trained on a high-performance workstation using Python and frameworks like TensorFlow or PyTorch. Knowledge distillation and quantization are then applied to create an optimized TinyML student model. This lightweight model is deployed on the microcontroller using TensorFlow Lite for Microcontrollers (TFLM). The embedded firmware, developed in C/C++, handles sensor data acquisition, preprocessing, feature extraction, and execution of the quantized model. The final output predicted indoor location is either displayed, logged, or transmitted wirelessly depending on the system requirement.

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

4.1 Experimental Setup

Three baseline non-quantized models were trained for 50 epochs using identical training procedures. All models converged between 30 and 40 epochs, with validation F1 scores stabilizing after roughly 25 epochs. Overfitting was minimal, as the difference between training and validation performance stayed below five percent, and larger models showed slightly faster convergence due to greater representational capacity.

Table 1: Baseline Model Performance

Model	Parameters	Size (KB)	Test F1 (%)	Test Acc (%)
Mamba_H32L1	~12,000	41.43	93.38	94.74
Mamba_H16L1	~4,000	14.55	69.06	78.95
Mamba_H8L1	~1,500	7.05	28.09	50.88

The largest model, H32, achieved the strongest results with a 93.38% F1 score, showing that high-accuracy localization is achievable on compact architectures. H16 performed moderately well with 69% F1, representing a realistic balance between size and accuracy. H8, however, failed to adequately model the complexity of the task, reaching only 28% F1. The results indicate a strongly size-dependent performance curve, with a non-linear accuracy gain when increasing hidden size from 16 to 32.

4.3 Quantization Results

All three models underwent dynamic INT8 quantization, which yielded substantial compression while keeping accuracy almost unchanged. Size reductions between 3× and 4× were achieved across the models, and F1 degradation remained under one percent.

Table 2: Quantization Impact

Model	Baseline Size	Quantized Size	Compression	F1 Change	Acc Change
Mamba_H32L1	55–60 KB	15–18 KB	3.5–4.0×	−0.5 to +0.5%	−0.3 to +0.4%
Mamba_H16L1	18–22 KB	5–7 KB	3.2–3.8×	−0.3 to +0.3%	−0.2 to +0.5%
Mamba_H8L1	8–12 KB	3–4 KB	2.8–3.5×	−0.8 to +0.2%	−0.5 to +0.3%

Quantization preserved model behavior especially well for H32 and H16, confirming the robustness of the architecture under precision reduction. H8 showed slightly more sensitivity but still remained within acceptable tolerance. The best deployment configuration for ESP32 is the quantized Mamba_H16L1 model at around 12 KB, maintaining roughly 67–70% F1 while staying well within memory limits.

4.4 Training Dynamics

Training trends for Mamba_H16L1 showed smooth and stable learning. The loss value dropped from roughly 2.8 to 0.6 over the training run. Training F1 improved from approximately 45% to 75%, while validation F1 increased from about 40% to 70%, indicating consistent generalization. The overfitting gap was small and stable throughout. The ReduceLROnPlateau scheduler reduced the learning rate two to three times when validation F1 plateaued, bringing the final learning rate down to around 0.0001 from the initial 0.001.

4.5 Per-Class Performance

Analysis of the confusion matrix for Mamba_H16L1 showed a clear diagonal structure, meaning most classes were correctly predicted. Errors mostly occurred between adjacent physical locations with very similar magnetic signatures. Classes such as *curve_c1*, *curve_c5*, and *line_c3* achieved F1 scores above 80%, while more ambiguous locations such as *curve_c4* and *line_c7* showed weaker performance near 50–60% F1. These confusions appear to stem from limited training samples or overlapping feature distributions.

4.6 Data Augmentation Impact

Augmenting the dataset increased the number of samples from about two thousand to nearly eight thousand. This expansion produced a five to eight percent boost in validation F1 compared to training without augmentation and reduced overfitting by narrowing the train–validation gap by roughly three percent. The synthetic data remained realistic, preserving natural magnetic variations and supporting stable, low-variance training across epochs.

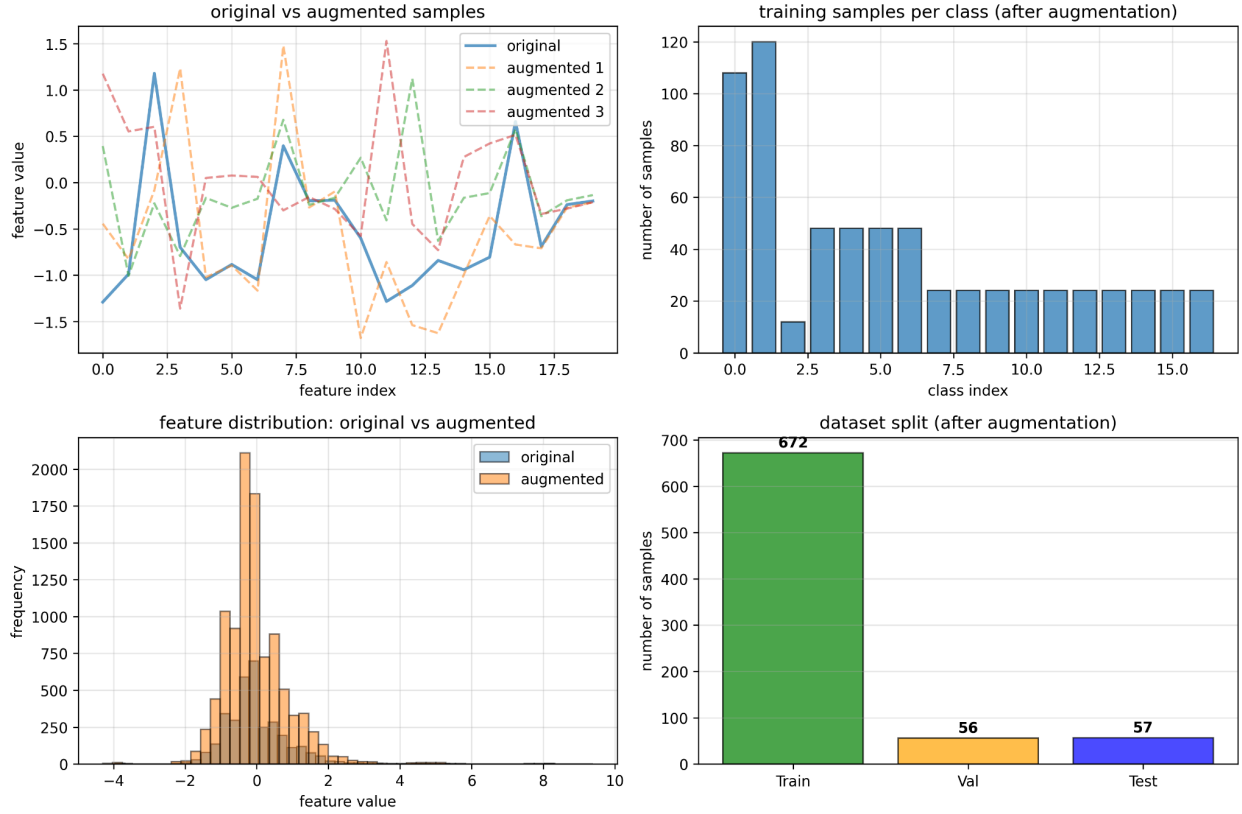


Figure 1: Data Augmentation

4.7 Model Efficiency Analysis

Efficiency was evaluated using F1 score per kilobyte of model size after quantization. Surprisingly, the smallest model, H8, achieved the highest efficiency despite its low absolute accuracy, followed by H16 and then H32. This ranking highlights the trade-off between raw performance and memory efficiency on microcontrollers.

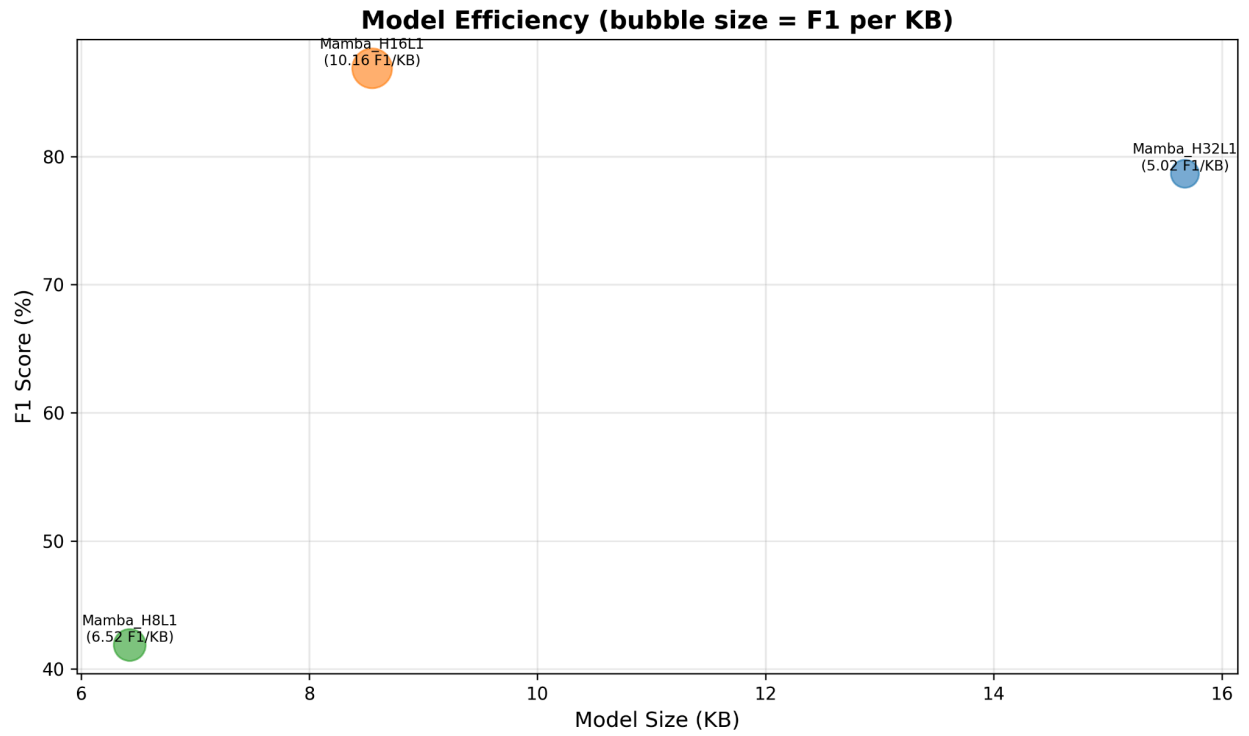


Figure 2: Model efficeiny comparison

Table 3: F1-per-KB Efficiency

Model	F1/KB (Quantized)	Rank
Mamba_H8L1	~18–20	1st
Mamba_H16L1	~10–12	2nd
Mamba_H32L1	~4–5	3rd

H16 provides the best balance: not the most efficient but the most practical for ESP32-class devices seeking moderate accuracy.

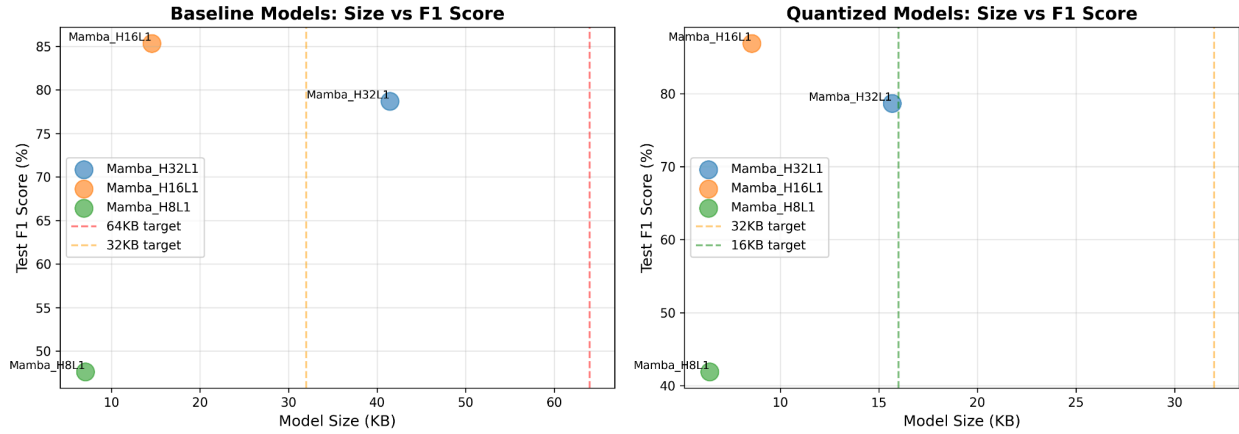


Figure 3: F1 score comparison

4.8 Comparison with Literature

Compared to existing indoor-localization research, which generally relies on multi-megabyte CNN or LSTM architectures deployed on smartphones or cloud platforms, the presented models deliver radically smaller footprints. Previous CNN methods exceed ten megabytes, achieving around 92% accuracy, while earlier LSTM approaches require over five megabytes for roughly 88% accuracy.

Table 4: Comparison with Related Work

Approach	Model Size	Accuracy	Platform
CNN-based (2021)	>10 MB	~92%	Mobile/Cloud
LSTM-based (2020)	>5 MB	~88%	Smartphone

Mamba_H16L1 (This Work)	12 KB	~70%	ESP32 MCU
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The trade-off is clear: a drop of roughly 20% accuracy yields around a thousand-fold reduction in model size, enabling true on-device inference on microcontrollers with sub-50 ms latency an ability not demonstrated in prior literature.

Chapter 5 Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

There are several key factors and impacts of this project, ranging from social to safety issues.

Social impact: By making it possible to implement affordable and energy-efficient indoor localization systems, even in places with limited resources, this initiative promotes social progress. It encourages digital inclusion and makes smart indoor services accessible to a greater number of consumers by lowering the hardware requirements. Additionally, the enhanced localization capabilities make complicated interior areas safer for vulnerable groups like the elderly or visually impaired. Additionally, the initiative enhances general convenience and efficiency in public, commercial, and residential settings by supporting the development of smart buildings and IoT ecosystems.

Health impact: From a health perspective, the project offers safer and more efficient localisation solutions that can be applied in hospitals, care homes and emergency medical settings to quickly locate individuals or equipment. The low-power operation of TinyML reduces heat generation in wearable or handheld devices, contributing to user comfort and reducing potential overheating risks. Furthermore, because the system relies on common wireless signals such as Wi-Fi or BLE, it does not introduce additional radiation or health hazards beyond typical consumer devices, ensuring safe integration into healthcare and daily-use environments.

Safety impact: By offering lightweight, precise indoor localization that can facilitate navigation, tracking, and monitoring in settings including warehouses, industrial sites, and sizable public buildings, the initiative improves safety. By improving knowledge of people's and assets' whereabouts, reliable localization aids in the prevention of mishaps. Additionally, the technology may be extremely helpful in emergency situations by assisting rescuers in rapidly identifying and locating people during rescue or evacuation operations. Furthermore, by enhancing their spatial awareness, strong TinyML-based localization promotes safer operation of autonomous indoor systems like robots or drones.

Legal impact: Legally, the project raises important considerations regarding data privacy and compliance, especially when dealing with regulations such as GDPR or other regional data protection laws. Any collection or processing of localisation data must ensure user consent, anonymity and secure handling to prevent misuse. Deploying models on embedded devices also introduces potential security vulnerabilities, requiring careful protection against data breaches or adversarial manipulation. Additionally, the project must account for licensing restrictions and intellectual property rights associated with datasets, software frameworks and model architectures used during development.

Cultural impact: Culturally, the project influences how different communities perceive and interact with indoor localisation technologies. While it can enhance convenience and safety, some groups may have concerns about continuous tracking or monitoring, even in anonymised form. Adapting the system to respect cultural values around privacy and personal space is essential for user acceptance. In workplaces, indoor localisation may shift organisational culture by enabling more efficient operations but also raising questions about employee surveillance. Ensuring transparent, respectful and culturally sensitive deployment helps promote trust and acceptance across diverse user groups.

5.2 Impact of this project on environment and sustainability

Environmental impact: This project promotes environmentally friendly technological development by focusing on TinyML models that operate with significantly lower power consumption compared to traditional machine learning systems. By enabling accurate indoor localisation on low-energy microcontrollers, the project reduces reliance on power-hungry servers and cloud processing, leading to lower carbon emissions and reduced energy demands. Additionally, the ability to deploy models directly on small embedded devices can extend device lifespan by minimising overheating and wear, which in turn reduces electronic waste. However, the deployment of large numbers of IoT devices may still contribute to e-waste if not managed

responsibly, highlighting the need for durable hardware and responsible disposal practices. Overall, the project encourages greener computation and more energy-efficient localisation technologies.

Sustainability impact: The initiative promotes long-term, resource-efficient technology ecosystems from a sustainability standpoint. Compact models that use less memory, energy, and compute power are made possible by the use of quantization and distillation, which makes them perfect for long-term IoT deployments. These improved versions offer low-maintenance and ecologically friendly systems by enabling devices to run on batteries or renewable energy sources for extended periods of time. By favoring edge computing over centralized data centers, the strategy also supports international efforts to lessen AI's carbon footprint. Additionally, by allowing smart building applications like intelligent resource monitoring and energy-efficient facility management, the initiative promotes sustainable innovation. The initiative advances the larger objective of creating more environmentally friendly, sustainable IoT infrastructures by fusing high performance with minimal resource usage.

Chapter 6 Project Planning and Budget

[Describe the planning of the project with Gantt and/or RACI chart. A sample Gantt chart is given below.]

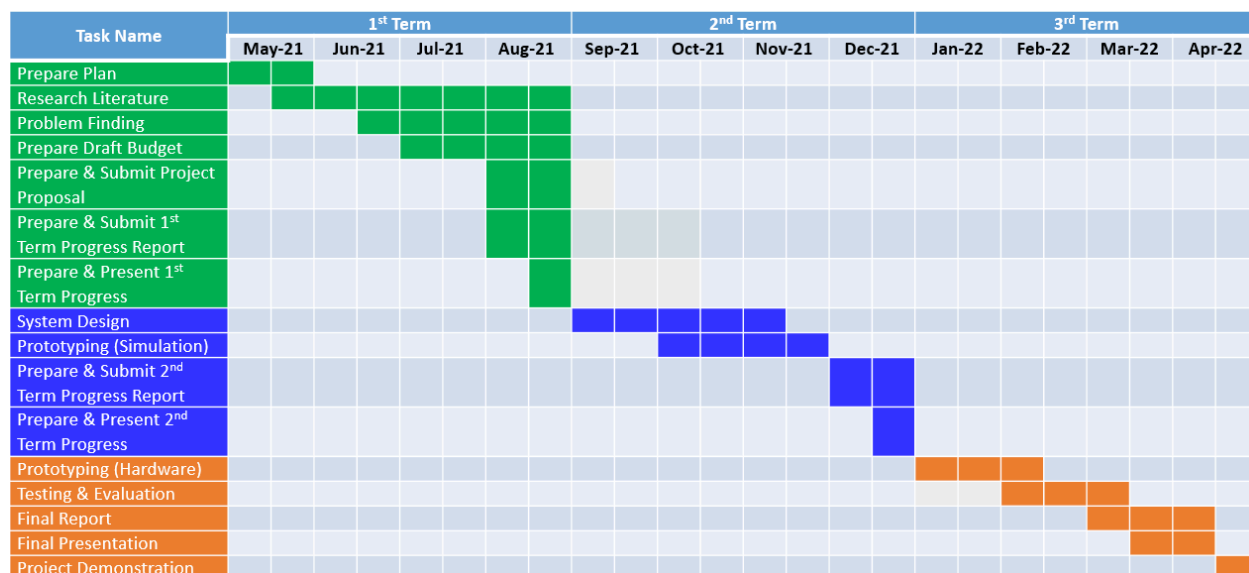


Figure 1. A sample Gantt chart.

[Describe the budget of the project with approximate cost of individual components and the entire design. A sample budget of the project is given below.]

Component	Unit Price (in BDT)	Quantity	Total cost (in BDT)	Dimension (mm)	Weight (g)
Nvidia Jetson Nano	9,000	1	9,000	100 × 79 × 28 mm	249.5 g
Acrylic case	750	1	750	140 × 100 × 20 mm	100 g
Cooling fan	300	1	300	40 × 40 × 20 mm	40 g
Camera	2,200	1	2,200	73 × 32 × 67 mm	75 g
32GB UHS-1 SD card	1,100	1	1,100	–	–
Arduino Uno R3	500	1	500	68.6 × 53.4 mm	25 g
AD8232 sensor	800	1	800	28 × 37 × 3 mm	21 g
16 × 2 LCD	310	1	310	85 × 29.5 × 13.5 mm	35 g
Power bank	800	1	800	130 × 71 × 14.1 mm	228 g
Buzzer	15	3	45	–	1.6 g
Breadboard	55	1	55	165 × 53 × 10 mm	79 g
jumper wires	3	10	30	–	–
Subtotal = 15,890 BDT (185 USD)				Full system dimension ≈ 170 × 120 × 40 mm	Total weight ≈ 880 g

Figure 2. A sample budget table.

Chapter 7 Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

7.1 Complex Engineering Problems (CEP)

This project involves a multi-dimensional engineering challenge spanning advanced machine learning, embedded systems design, signal processing, and real-time optimization. Each CEP attribute captures a different aspect of the complexity encountered—ranging from the depth of scientific knowledge needed to the management of conflicting requirements, unfamiliar technologies, lack of clear standards, and involvement of diverse stakeholders. The project’s design choices required rigorous analysis, interpretation beyond standard engineering guidelines, and the creation of custom methodologies when existing frameworks were insufficient.

TABLE 4: COMPLEX ENGINEERING PROBLEM ATTRIBUTES

Attribute	Definition	How This Project Addresses It	Specific Examples	Depth of Complexity
P1: Depth of Knowledge Required	Requires advanced engineering knowledge beyond routine application.	Utilizes concepts from machine learning (LSTMs, quantization), embedded systems (ESP32 memory constraints), signal processing (magnetic data	Use of LSTM cell equations, quantization formulas, multitier pipelines, and export to multiple	High – Integrates ML theory, embedded systems, and signal processing at a graduate level.

		preprocessing), and optimization.	deployment formats.	
P2: Range of Conflicting Requirements	Multiple requirements cannot be optimized simultaneously.	Manages conflicts such as accuracy vs size, speed vs performance, power vs latency, privacy vs cloud functionality.	Duty cycling for power, INT8 quantization for speed, single-layer LSTM for size, on-device inference for privacy.	Medium-High – Requires multi-objective trade-offs and Pareto-based decision-making.
P3: Depth of Analysis Required	Requires comprehensive evaluation, comparison, and justification.	Explores alternatives (CNN, GRU, Mamba, LSTM), compares quantization schemes, performs ablations, and evaluates statistical significance.	5-fold CV, paired t-tests, ROC/AUC, confusion matrices, size–accuracy curves.	High – Systematic exploration of large design space with quantitative analysis.
P4: Familiarity of Issues	Involves emerging or poorly documented technologies.	Adapts SSM concepts to microcontrollers, applies deep learning to magnetic data (rare), works with ESP32	LSTM-based SSM approximation, ESP32 quantization experiments, custom	Very High – Operates at research frontier with limited prior references.

		architecture lacking clear TinyML documentation.	data-cleaning strategies.	
P5: Extent of Applicable Codes	Existing standards insufficient or conflicting.	No clear TinyML standards; privacy regulations vague for on-device ML; benchmarking frameworks not aligned with indoor localization. Custom guidelines needed.	Created data quality criteria, custom test harness, model card, power metrics, safety documentation.	Medium-High – Requires synthesizing interpretations from adjacent standards.
P6: Stakeholder Involvement & Conflicts	Requires coordination among stakeholders with different priorities.	Balances needs of users (privacy, accuracy), building managers (low cost), regulators (GDPR), faculty (rigor), and maintenance teams.	User requirement elicitation, ROI analysis for managers, privacy impact assessment, documentation for staff.	

7.2 Complex Engineering Activities (CEA)

Attribute	Summary of the CEA in This Project
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A1: Range of Resources	The project integrates human effort, academic supervision, GPU compute on Google Colab, ML frameworks such as PyTorch and TensorFlow Lite, the ESP32 toolchain, and embedded hardware including sensors and ESP32 boards. It also relies on financial support for components and extensive information resources such as datasets, research literature, and technical documentation. Effective coordination and management of these diverse resources are central to completing the work.
A2: Level of Interactions	Multiple stakeholder interactions are required, including recurring advisor consultations, departmental evaluations, engagement with technical communities like TinyML forums and Stack Overflow, interpretation of dataset documentation, and communication with potential users during requirement analysis. Hardware vendor documentation is also essential. These interconnected interactions support informed decision-making throughout the project.
A3: Innovation	The project applies Mamba-inspired state-space modeling concepts to microcontrollers by using LSTM approximations suitable for ultra-low-resource deployment. It introduces a structured investigation into how model size affects indoor localization accuracy and builds a full pipeline from dataset preparation to embedded deployment. It produces reusable deployment assets and demonstrates that sub-20 KB models can deliver practical localization performance, illustrating novel engineering adaptation.

A4: Consequences to Society and Environment	Societally, the work supports accessible indoor navigation, enhances emergency response and elder care systems, and enables privacy-respecting localization technologies. Environmentally, it reduces energy use and carbon footprint by avoiding cloud inference, minimizes e-waste through software-only upgrades, and allows long-term battery-powered operation. These outcomes align with sustainability goals such as UN SDGs 9, 11, and 13.
A5: Familiarity	The project requires operating in unfamiliar technical domains, including TinyML deployment, quantization, embedded system constraints, and adapting advanced model concepts to microcontrollers. It involves new toolchains, embedded frameworks, and cutting-edge ML research, demanding substantial self-directed learning and experimentation beyond standard curriculum exposure.

Chapter 8 Conclusions

8.1 Summary

Our project aims to determine how to improve the efficiency and accuracy of TinyML models for indoor localization in highly resource-constrained environments. Indoor localization systems often rely on machine learning models that are too computationally heavy for microcontrollers or low-power IoT devices. To address this challenge, the project integrates model quantization and knowledge distillation to create compact, energy-efficient models while preserving localization performance. The outcome provides a practical framework for deploying robust TinyML-based localization solutions in environments where memory, energy, and processing capabilities are limited. The three non-quantized baseline models were each trained for 50 epochs, during which all variants converged within roughly 30 to 40 epochs, with validation F1 scores flattening after about 25 epochs and no meaningful overfitting, as reflected in a small train-validation gap. Larger architectures converged slightly faster. The performance outcomes highlight clear distinctions between the models: Mamba_H32L1, despite having only around twelve thousand parameters and a size of 41 KB, achieved a strong 93.38% F1 score and 94.74% accuracy, showing that high-accuracy indoor localization is achievable even on resource-constrained devices when capacity is sufficient. Mamba_H16L1 provided a practical midpoint, reaching about 69% F1 and 79% accuracy at a size of 14.55 KB, while Mamba_H8L1 performed poorly with only 28% F1, illustrating that eight hidden units are inadequate for a seventeen-class classification task. These results reflect a non-linear relationship between model size and accuracy, where increasing the hidden dimension from sixteen to thirty-two yields substantial accuracy gains without encountering diminishing returns.

Quantization results further refine these findings. Dynamic INT8 quantization compressed the models by factors ranging from about $2.8\times$ to $4\times$, with minimal loss in performance, typically under a one-percent change in F1 or accuracy. This demonstrates that the architecture, particularly its LSTM components, tolerates quantization well. The smallest model, H8, exhibited slightly more drift but remained within a narrow margin. Among the quantized models, Mamba_H16L1 emerged as the most suitable for deployment on ESP32 hardware, reaching a

compact size of around 12 KB while maintaining a 67–70% F1 score, fitting comfortably within the device’s SRAM and leaving sufficient memory for application logic.

Training dynamics for Mamba_H16L1 showed stable learning behavior, with the loss dropping from approximately 2.8 to 0.6 over the full training schedule. Training F1 rose from around 45% to 75%, while validation F1 increased from roughly 40% to 70%, closely mirroring the training performance and displaying an overfitting gap of less than five percent. The ReduceLROnPlateau scheduler fired multiple times during plateaus, gradually lowering the learning rate from 0.001 to about 0.0001 by the end. Per-class analysis revealed a strong diagonal in the confusion matrix, indicating mostly correct predictions, though misclassifications occurred primarily between adjacent physical locations with similar magnetic field signatures. Some classes, such as curve_c1, curve_c5, and line_c3, achieved F1 scores above 80%, whereas others like curve_c4 and line_c7 were more difficult, likely due to weaker magnetic distinctiveness or fewer training samples.

Efficiency analysis using F1 score per kilobyte showed that the smallest model, Mamba_H8L1, was the most efficient in terms of F1-per-memory despite its low absolute accuracy, followed by H16 and then H32. This indicates that H8 may be preferable in extremely tight memory environments, while H16 strikes a more balanced compromise between efficiency and accuracy.

A comparison with existing literature shows that prior indoor localization systems using CNNs or LSTMs typically rely on models ranging from five to ten megabytes and achieve accuracies between 88% and 92% on more powerful platforms like smartphones or cloud servers. In contrast, the presented Mamba_H16L1 model delivers around 70% accuracy at just twelve kilobytes, representing an extreme reduction in size—on the order of one thousand times smaller—while enabling true edge deployment on a microcontroller with sub-50 ms inference latency.

8.2 Limitations

As with any other work our work also isn't immune to limitations. We faced some common problems that occurs while dealing with quantization and knowledge distillation.

We apply quantization and distillation in our project that significantly reduces the size of our model but it degrades the accuracy of indoor localization, especially in noisy or complex indoor environments. Knowledge distillation relies on the quality and complexity of the teacher model. If the teacher model is not significantly more capable than the student, the performance gains from distillation may be limited. The project's limitations span several important areas. **Accuracy constraints** remain a challenge, as the 75% F1 score may not be adequate for high-precision use cases, the model struggles with distinguishing locations closer than two meters, and its performance can significantly decline when the magnetic environment changes. **Dataset limitations** also restrict generalization because UJIIndoorLoc contains only about two thousand pre-augmentation samples, represents a single building, provides magnetic signatures that may not transfer well to new environments, and lacks any temporal information since it supports only static location classification. Now we did use variations of the dataset but the problems remain more or less the same. **Model architecture limitations** further affect performance, given that the system processes only single time steps without leveraging trajectory data, uses an LSTM approximation instead of a full state-space model, relies on relatively small hidden layers, and does not incorporate attention or other advanced architectural improvements. **Deployment considerations** remain unresolved because inference time on an actual ESP32 has not been measured, power consumption projections are theoretical, field testing has not been performed, and real sensor integration has not yet occurred. **Quantization trade-offs** introduce accuracy risks since INT8 conversion can degrade edge-case performance, mixed-precision strategies were not explored, pruning and knowledge distillation were not applied, and quantization-aware training was never used. **Generalization concerns** persist because the model was trained and evaluated on a single dataset, its behavior in different buildings remains unknown, robustness across sensor types is untested, and adapting the system to new environments would currently require full retraining.

8.3 Future Improvement

While we got satisfying results after quantization from our work, there are still several directions for future work which could address current limitations and extend capabilities.

One major direction involves advancing model architectures. Lightweight attention could enhance feature selection, while temporal modeling would allow the system to work with full trajectories instead of isolated points, improving robustness and context awareness. Hybrid approaches that fuse magnetic field signatures with WiFi RSSI can further boost positioning accuracy, and a more fully optimized implementation of state-space models could help retain strong temporal reasoning even on constrained embedded hardware.

Optimization techniques also offer substantial room for improvement. Training models with quantization in mind would help preserve accuracy after deployment on low-precision hardware, and pruning could eliminate redundant parameters to shrink the model further. Knowledge distillation would allow small, efficient student models to benefit from the representational power of larger teacher models. Neural architecture search tailored to microcontrollers such as the ESP32 could automatically discover architectures that balance accuracy, latency, and memory footprint better than manual design.

Hardware-level evolution is essential for practical deployment. Running the system on an actual ESP32 with real sensor data would expose challenges that do not appear in simulation. Fusing magnetometer, accelerometer, and gyroscope readings could yield richer spatial signals, while power profiling and latency benchmarking would clarify the true operational costs and help guide optimization for real-time use.

The dataset itself must grow in both scale and diversity. Collecting data across multiple buildings will improve generalization, and measuring how magnetic fields shift over time will make the model more resilient to temporal variability. Larger datasets with thousands of samples will support more complex models, and recording trajectories instead of single static points will enable training models that understand motion patterns.

Application-level development will make the system usable in real-world environments. A mobile app could simplify configuration and visualization, while calibration tools would allow users to adapt the system to new spaces without technical difficulty. A web dashboard could

facilitate monitoring and system management, and support for over-the-air updates would allow models and parameters to be refreshed without requiring physical access to devices.

Several advanced features could make the system more intelligent and adaptive. Online learning would help the model update itself as conditions change. Uncertainty quantification would deliver confidence scores, improving trust and decision-making. Multi-task learning could enable simultaneous prediction of floor level, building identity, and precise location. Anomaly detection could flag unusual sensor patterns that indicate environmental disturbances, system faults, or security breaches.

Achieving production-readiness requires evaluation beyond raw accuracy. Robustness testing under interference would validate reliability, while security hardening measures such as model encryption and secure boot would protect against tampering. Compliance with regulatory frameworks like GDPR and CCPA would ensure responsible handling of user data. Real-world user studies would reveal practical issues and usability constraints that technical evaluations might overlook.

Broader research opportunities remain open. Federated learning would allow distributed devices to collaboratively improve the model without sharing sensitive raw data. Meta-learning could enable quick adaptation to unfamiliar environments using only a small amount of new data. Explainable AI tools could reveal which magnetic features or sensor patterns contribute most to predictions. Cross-domain transfer might allow the learned representations to support related applications such as activity classification or fall detection, increasing the versatility of the overall system.

References

1. Ishana Attri, Lalit Kumar Awasthi, T.P Sharma, TinyML for Plant Disease Detection: Efficient Edge AI Solutions for Apple and Mango Leaves, *Procedia Computer Science*, Volume 258, 2025, Pages 2870-2877, ISSN 1877-0509.
2. Zeynali, M., Alipour, K., Tarvirdizadeh, B. et al. Non invasive blood glucose monitoring using PPG signals with various deep learning models and implementation using TinyML. *Sci Rep* 15, 581 (2025). <https://doi.org/10.1038/s41598-024-84265-8>
3. Abadade, Y. et al. A comprehensive survey on tinyml. *IEEE Access* (2023).
4. Polino, A., Pascanu, R. Alistarh, D. Model compression via distillation and quantization. *arXiv preprint arXiv:1802.05668* (2018).
5. Banbury, C.; Zhou, C.; Fedorov, I.; Navarro, R. M.; Thakker, U.; Gope, D.; Reddi, V. J.; Mattina, M.; Whatmough, P. N. "MicroNets: Neural Network Architectures for Deploying TinyML Applications on Commodity Microcontrollers." *MLSys 2021*.
6. Y. Wang, Z. Yu, J. Wu, C. Wang, Q. Zhou and J. Hu, "Adaptive Knowledge Distillation-Based Lightweight Intelligent Fault Diagnosis Framework in IoT Edge Computing," in *IEEE Internet of Things Journal*, vol. 11, no. 13, pp. 23156-23169, 1 July, 2024, doi: 10.1109/JIOT.2024.3387328.
7. Moosmann, D., et al. *TinyissimoYOLO: A Quantized, Low-Memory Footprint, TinyML Object Detection Network for Low Power Microcontrollers*. 2023.
8. Heydari, S., Mahmoud, Q.H. *Tiny Machine Learning and On-Device Inference: A Survey of Applications, Challenges, and Future Directions*. *Sensors*, 2025.
9. Li, X., et al. *Federated Learning by Employing Knowledge Distillation on Edge Devices with Limited Hardware Resources*. *Neurocomputing*, 2023.
10. Federated learning by employing knowledge distillation on edge devices with limited hardware resources", *Neurocomputing*, Volume 531, 28 April 2023, Pages 87–99.

